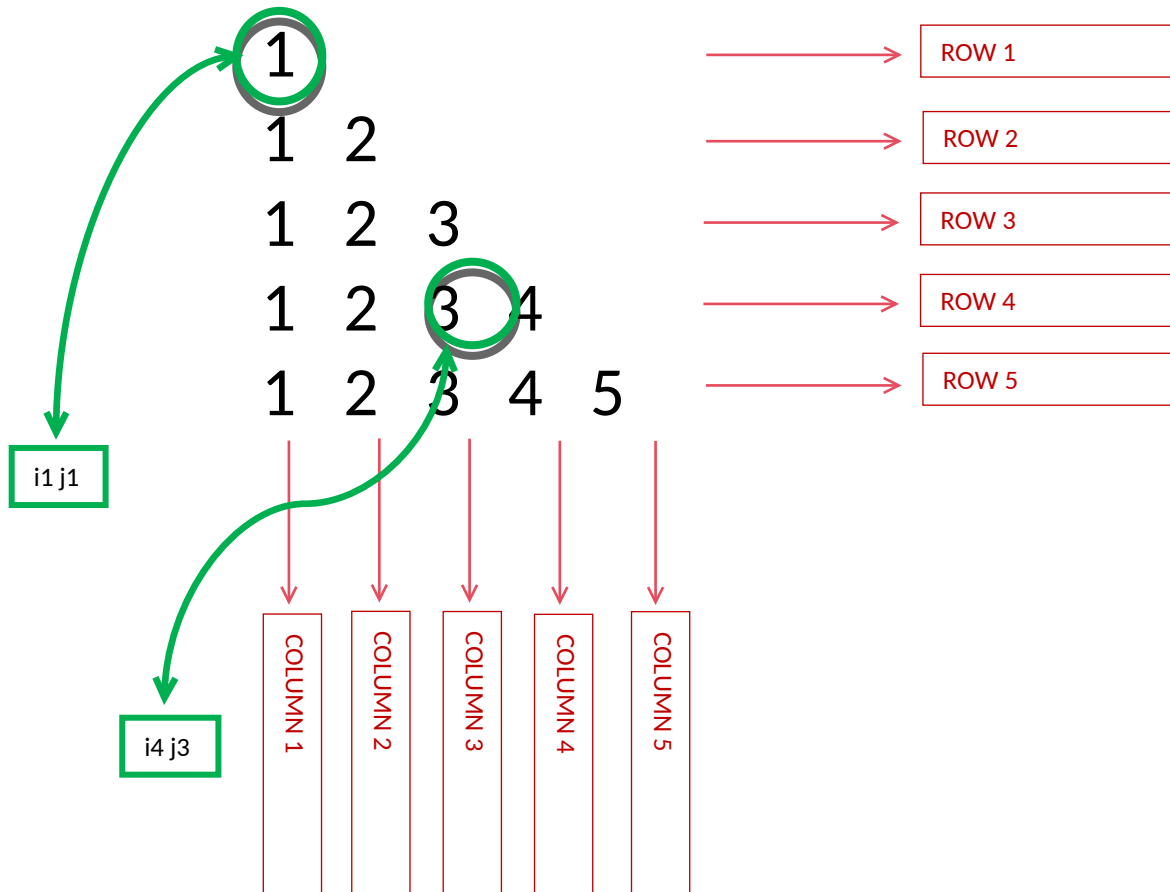


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

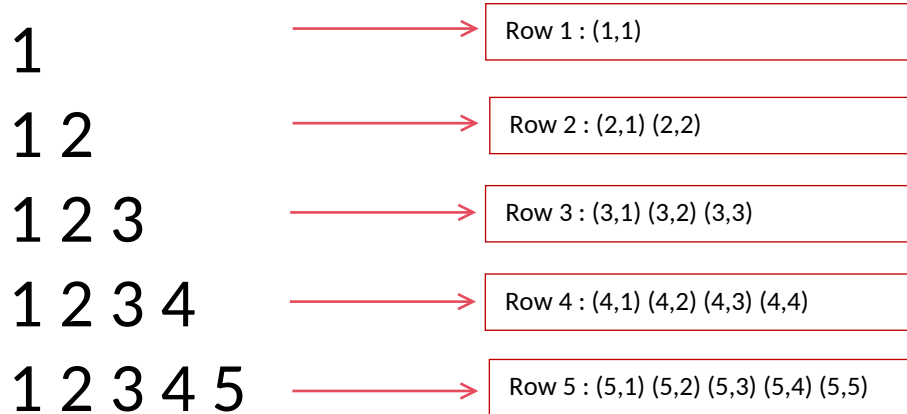
1. NUMBER TRIANGLE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a **right-angled number triangle shape**.
- This pattern is based on the **Right Angled Triangle structure**.
- Unlike star patterns, this pattern uses **numbers instead of symbols**.
- The number of elements increases row by row.
- Each row starts from 1 and goes up to the current row number.
- For $n=5$



i. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a triangular portion of an $n \times n$ grid. Each row prints only a subset of the available columns.

- size = $n \times n$.

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

- For $n = 5$:

The number in each row depends directly on the row number/index

Observation:

- Row 1 → 1
-
- Row 2 → 1 2
-
- Row 3 → 1 2 3
-
- Row 4 → 1 2 3 4
-
- Row 5 → 1 2 3 4 5
-

A clear relationship can be observed: Number of elements = row index

Step 2: Traverse the grid row by row using outer loop.

- The outer loop is responsible for generating rows.

- for (i = 1 to n)

Step 3: For every row, traverse columns .

- for (j = 1 to i)

Step 4: Print value = j

Step 5: After all numbers of the current row are printed, move to the next line.

Step 6: Repeat the same process until all n rows have been completed.

- The pattern growth is controlled by the inner loop limit.

- the inner loop runs from 1 to the current row number:

j <= i (This automatically increases the number printed in each row.)

{

for(j = 1; j <= i; j++)

Meaning:

Current row gets as many number as its row number.

}

2. Program :-

```
package week1_module;
import java.util.Scanner;

public class NumberTriangle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

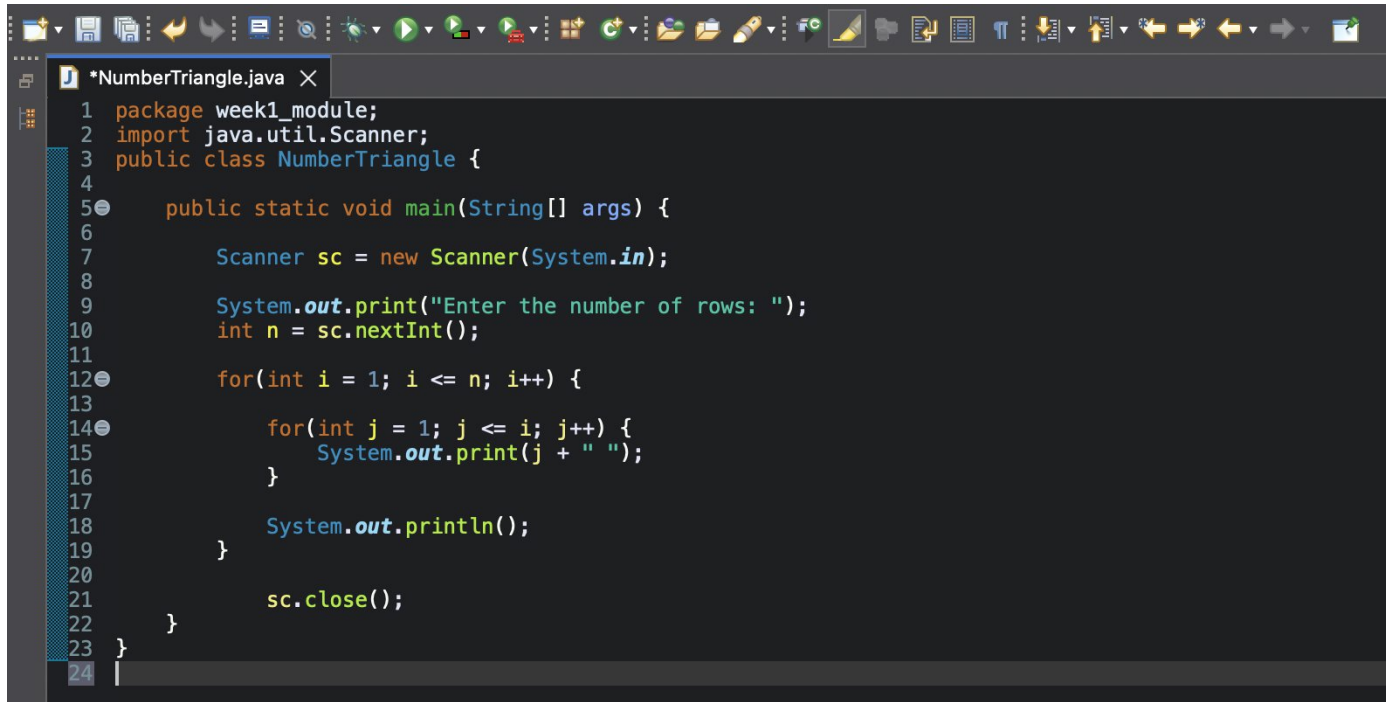
3. Program Output :-

Enter the size of the rows: 5

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

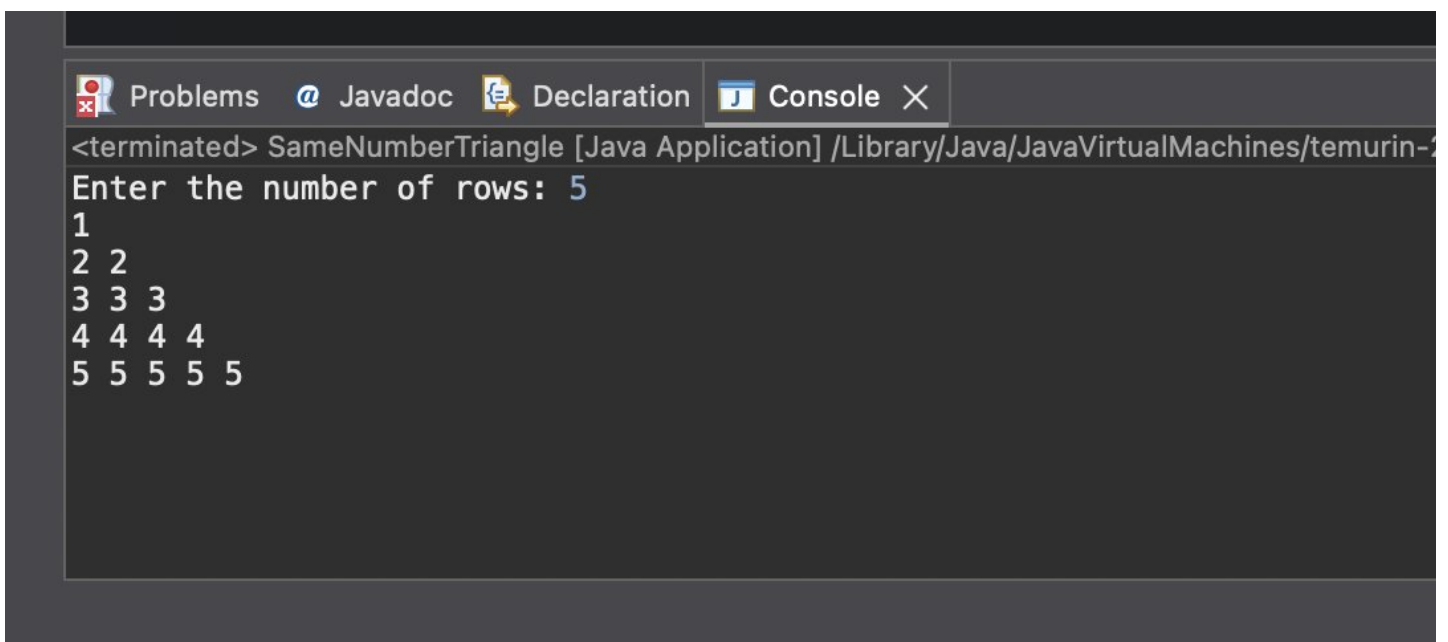
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class NumberTriangle {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the number of rows: ");
10        int n = sc.nextInt();
11
12        for(int i = 1; i <= n; i++) {
13
14            for(int j = 1; j <= i; j++) {
15                System.out.print(j + " ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
```

ii. Program's Output :-



```
<terminated> SameNumberTriangle [Java Application] /Library/Java/JavaVirtualMachines/temurin-2
Enter the number of rows: 5
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```